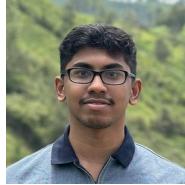


S K Hari Keshav

Project Research Associate · RISE Lab, IIT Madras



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harikheshav.github.io/portfolio
Chennai, India

EDUCATION

MS by Research (Project Mode), Computer Science and Engineering

Indian Institute of Technology Madras

Jul 2024 – Present

CGPA: 7.67/10

Coursework: Computer Architecture, CAD for VLSI Systems, Digital System Testing & Testable Design, Security in Cyber-Physical Systems, Systems Engineering for Deep Learning, Recent Topics in IoT-based Sensing

Bachelor of Engineering (B.E.), Electronics and Communication Engineering

Sri Venkateswara College of Engineering, Sriperumbudur

2018 – 2022

CGPA: 7.77/10

Senior Secondary (XII), Science – CBSE

Kola Saraswati Vaishnav Chetty Senior Secondary School

Year of completion: 2018 · Percentage: 79.20%

Secondary (X) – CBSE

Sindhi Model Senior Secondary School

Year of completion: 2016 · CGPA: 8.20/10

EXPERIENCE

Project Research Associate – Shakti Processor

RISE Lab, Indian Institute of Technology Madras, Chennai

Oct 2023 – Present

- Parameterised AXI4 ID widths across SoC peripherals (UART, GPIO, DMA, QSPI, CLINT, PLIC) and the AES, RSA and SHA crypto blocks, in Bluespec SystemVerilog.
- Adding an APB bus to the Shakti fabrics, with an APB error slave and SoC-level compatibility tests.
- Contributed to the C-Class core's instruction-trace logic and configuration, along with trace scripts for verification.
- Test the designs and develop their driver code in C.

Associate Software Engineer

Eurofins IT Solutions India Pvt Ltd, Chennai

Jul 2022 – Jul 2023

- Built web applications and APIs with C#, ASP.NET and MS SQL Server, front ends in Angular and TypeScript, and worked on Dynamics AX.

INTERNSHIPS

Research Intern – Automated Weed Control by a Swarm of Robots

Sri Venkateswara College of Engineering (externally sponsored project)

Nov 2021 – Jun 2022

One of four engineers building a minimum viable prototype robot; built the

TensorFlow model and the hardware prototype that uses it for autonomous movement.

Web and Software Development Intern

Rajalle Container Movers Pvt Ltd (remote)

May 2021 – Jul 2021

github.com/Harikheshav/RCM_Specifics_2.0

Built desktop (Tkinter) and web (Django) apps for vehicle-movement data entry with suggestion-based filtering, plus a vehicle expense tracking system.

PROJECTS

CNN Accelerator on a Shakti RISC-V SoC – Drowsiness Detection

2026 · CS6651 course project

github.com/Harikheshav/Drowsiness-Detection

Parameterised Bluespec convolution engine (configurable MAC lanes, int8 data with int32 accumulation) integrated into the Shakti SoC on a Nexys Video FPGA. Models compiled from ONNX with TVM to C; accelerator and CPU-only builds validated against each other on hardware via OpenOCD/GDB.

Leela Chess Zero on a RISC-V SoC with Gemmini

2025 · CS6886 course project

github.com/Harikheshav/CS6886_CourseProject

Cross-compiled the Leela Chess Zero neural-network engine for RISC-V and built a Chipyard SoC with the Gemmini systolic-array accelerator for the Xilinx VC707 FPGA, with Linux workloads built through FireMarshal.

SKILLS

Hardware & architecture

Bluespec SystemVerilog, Verilog, RISC-V, AXI4 / APB, SoC peripherals, design for testability

ML systems

PyTorch, TensorFlow, TVM, ONNX

Languages

Tamil (native), English (full professional), Hindi (limited working)

Tools & platforms

Chipyard, Gemmini, Shakti, Vivado, Verilator, OpenOCD / GDB

Software

C / Embedded C, C++, Python, C# / .NET, SQL Server, Django, Angular, TypeScript

LINKS

Portfolio harikheshav.github.io/portfolio

GitHub github.com/Harikheshav

GitLab gitlab.com/harikheshav

LinkedIn linkedin.com/in/hari-kheshav-s-k-464154171

ACHIEVEMENTS

Won Idea Jam 2.0, conducted by the Department of CSE, Sri Venkateswara College of Engineering.

Secured second prize in Digital Simulation (D-SIM) at Invente 4.0, SSN College of Engineering, 2019.

Cleared the first three levels of the Swadeshi Microprocessor Challenge.